

The above equations gives

$$\begin{aligned}
\bar{w}_M^{(1)} &= \frac{1}{n} \sum_{i=1}^n x_1^{(i)} + \sum_{i=n+1}^M \frac{1}{n} (1 - \eta_0)^{i-n} x_1^{(i)} + \sum_{i=1}^n (1 - \eta_0)^{M-i} w_0^{(0)} \\
&= \frac{1}{n} \sum_{i=1}^n x_1^{(i)} + \sum_{i=n+1}^{n+1+\lfloor \log(Md) \rfloor} \frac{1}{n} (1 - \eta_0)^{i-n} x_1^{(i)} + \tilde{o}(d) + o(d) \\
&\leq \frac{1}{n} \sum_{i=1}^{n+1+\lfloor \log(Md) \rfloor} x_1^{(i)} (1 + o(1)).
\end{aligned} \tag{4}$$

The second equation uses the fact $n = \Theta(M^{\frac{2}{3}})$. Thus, we give the upper bound for the component of expected loss on the x-axis

$$\mathbb{E}[w_M^{(1)} w_M^{(1)}] = \tilde{O}(n^2 d^2) = \tilde{O}\left(M^{-\frac{2}{3}} L^2\right).$$

Similarly, for the y-axis, we have

$$\bar{w}_M^{(2)} \leq \frac{1}{n} \sum_{i=1}^{n+\log(Md)} x_i^{(2)} (1 + o(1)). \tag{5}$$

Notice that the only difference between the derivation of Equation (5) and the derivation of Equation (4) is we cannot replace α with α_0 since $x_1^{(2)}$ is not the constant 0, but the difference $(\alpha_0 - \alpha)w_M$ can be obviously merged into the main term $\frac{1}{n} \sum_{i=1}^{n+\log(Md)} x_i^{(2)}$. And then we get

$$\mathbb{E}[w_M^{(2)} w_M^{(2)}] = \tilde{O}\left(\frac{L^2}{n}\right) = \tilde{O}\left(M^{-\frac{2}{3}} L^2\right).$$

Finally, notice that the rescale constant α can be merged into the $\tilde{O}$ notation, thus we complete the proof. $\square$